

A categorical interpretation of possibilistic operators in formal concept analysis

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Abstract

In this short paper we describe Dubois–Prade’s eight possibilistic operators in formal concept analysis as arising from Kan extensions and complement duals of the underlying boolean profunctor. This perspective neatly uncovers the underlying reason for the proof that Π -pairs form the formal concepts of the complement context. We prove that the Galois connection and the Π -pairs are the only symmetric or asymmetric operator compositions that give formal concepts. We also consider categorical tools for constructing new types for concepts not previously studied in the FCA literature.

1 Introduction

Formal concept analysis (FCA) is a technique for structuring data by associated attributes [cite](#), and has during recent years seen a wide array of applications in various contexts [cite](#). Given a list of objects, a list of attributes, and a relation describing which object has which attribute, FCA constructs a set of *formal concepts* which are naturally structured into a lattice.

Formal concepts are constructed as fixedpoints of derivation operations $(-)'$ on the set of objects and attributes. More formally, given a set of objects G , a set of attributes M and a relation $R \subseteq G \times M$, one defines for a subset of objects $A \subseteq G$ its *intent*

$$A' = \{m \in M \mid \forall a \in A, aRm\},$$

and similarly for a subset of attributes $B \subseteq M$ its *extent*

$$B' = \{g \in G \mid \forall b \in B, gRb\},$$

where xRy means that the object x has the attribute y . In words, the extent of A is the set of common attributes shared by all of its objects, and the intent of

B is the set of objects that have all of its attributes. A formal concept is then a subset $(A, B) \subseteq G \times M$ such that $B = A'$ and $A = B'$, see [cite](#) for details.

In [cite](#), the authors take inspiration from possibility theory, see [cite](#), and introduce additional operators on G and M based on the possibilistic powerset operations introduced by Dubois–Prade in [cite](#). Denoting the previous FCA operator by $(-)' = (-)^\Delta$, they further introduce for a subset $A \subseteq G$ the operators

$$A^\Pi = \{m \in M \mid \exists a \in A, aRm\},$$

as well as

$$A^\nabla = \{m \in M \mid \exists a \notin A, a\bar{R}m\},$$

where $\bar{R} = \neg R$ is the negated relation, and finally

$$A^N = \{m \in M \mid \forall g \in G, gRm \implies a \in A\}.$$

These are usually called the potential possibility Π , the guaranteed necessity N , the guaranteed possibility Δ and the potential necessity ∇ . One can similarly define these four operators for subsets $B \subseteq M$, which we denote with subscripts, $(-)_{\Pi}, (-)_N, (-)_{\Delta}$ and $(-)_{\nabla}$.

By using a well established categorical framework for FCA, the first goal of this paper is to prove that the above eight operators arise canonically from the underlying relation R , rather than from possibility-theoretic considerations. More precisely, the relation R and its complement relation $\bar{R} = \neg R$ both give rise to four operations each via right and left Kan extensions, which together form four adjoint pairs on $P(G)$ and $P(M)$. This categorical viewpoint makes the properties of these operators, as studied in [cite](#), evident from general abstract nonsense.

By design, fixedpoints of $(-)_{\Delta} \circ (-)^{\Delta} : P(G) \longrightarrow P(G)$ are exactly the extents of the formal concepts of the formal context (G, M, R) . In [cite](#), the authors prove that

the fixedpoints for the asymmetric operator composition $(-)_\Pi \circ (-)^N$, which they call *NII*-pairs, are precisely the formal concepts of the complementary formal context $(G, M, \bar{R})$ defined using the negated relation $\bar{R}$. Via the categorical viewpoint, this result is given its *raison de etre*.

The authors in [cite](#) note at the end of the paper that further research should be done to understand the remaining asymmetric operator compositions of the possibilistic operators. We prove that the operators $(-)_\Delta \circ (-)^\Delta$ and $(-)_\Pi \circ (-)^N$ are the only ones that has formal concepts as their fixed points. Along the way we provide some simple categorical tools for studying such compositions, as well as other endofunctors on $P(G)$. In particular, any composition-sequence of the possibilistic left adjoints has a corresponding closure operator, giving a plethora of new types of “formal” concepts one can study.

2 Category theory and FCA

Formal concept analysis has attracted interest by category theorists for being a natural and useful application of their abstract tools. Category theory was introduced by Eilenberg and MacLane in [cite](#), and is an abstract mathematical theory for understanding compositional structure – see [cite](#) for a comprehensive introduction.

A category consists of *objects* and *morphisms*, which are abstract relationships between objects. Category theory can then be thought of as studying objects by understanding how they relate to all other objects of the same defined type. We will assume that the reader is familiar with basic categorical constructions.

For a category theorist, the objects G and attributes M can be viewed as discrete categories. Any relation can be described as a $\mathbb{2} = \{0, 1\}$ -valued profunctor, also called *boolean* profunctors. In our case the profunctor $R: G \nrightarrow M$ is a functor $G^{\text{op}} \times M \rightarrow \mathbb{2}$, which takes a pair (g, m) to 1 if g has the attribute m – see [cite](#) for details. This will be our running definition of a formal context.

Definition 2.1. A *formal context* (G, M, R) consists of two discrete categories G and M and a boolean profunctor $R: G \nrightarrow M$. We will say an object $g \in G$ has attribute $m \in M$ if $R(g, m) = 1$, usually written gRm .

From this perspective we can also produce the formal concepts associated to any formal context.

2.1 The closure operators

Fix a formal context (G, M, R) . To construct the associated formal concepts, notice that a the profunctor R is by currying equivalent to two functors

$$R^-: G \rightarrow [M, \mathbb{2}]^{\text{op}} \text{ and } R_-: M \rightarrow [G^{\text{op}}, \mathbb{2}].$$

We also have two Yoneda embeddings, giving a diagram

$$\begin{array}{ccc} G & \xrightarrow{R^-} & [M, \mathbb{2}]^{\text{op}} \\ y \downarrow & & \uparrow cy \\ [G^{\text{op}}, \mathbb{2}] & \xleftarrow{R_-} & M \end{array}$$

where y is the Yoneda embedding and cy is the Yoneda coembedding into the presheaf and opcopresheaf categories respectively. As G and M are discrete, the functor categories are precisely the powersets of G and M respectively: $[G^{\text{op}}, \mathbb{2}] = P(G)$ and $[M, \mathbb{2}]^{\text{op}} = P(M)^{\text{op}}$.

As $[G^{\text{op}}, \mathbb{2}]$ is complete, there is a unique colimit preserving right Kan extension R^* of R^- along the Yoneda embedding y , which we claim is the standard FCA closure on $P(G)$.

Lemma 2.2. *The right Kan extension R^* of R^- is the standard FCA closure operator $(-)^\Delta$.*

Proof. The functor R^* is given explicitly by using a categorical end-formula, which for a presheaf $A \in [G^{\text{op}}, \mathbb{2}]$ is

$$R^*(A)(m) = \int_{g \in G} \mathbb{2}(A(g), gRm).$$

In $\mathbb{2}$, the internal hom is given by *implication*, and the above end-formula is computed pointwise, reducing it to the conjunction

$$R^*(A)(m) = \bigwedge_{g \in G} (A(g) \implies gRm).$$

In other words, having $m \in R^*(A)$ is equivalent to having that for all g , if $g \in A$ then g has the attribute m . This is precisely the FCA closure operation $(-)^\Delta = (-)'$ on $P(G)$ as defined earlier. $\square$

Similarly, the other curried functor R_- has a right Kan extension R_* along the coYoneda embedding

$cY : M \longrightarrow [M, 2]^{\text{op}}$, which we claim is the other standard FCA closure operation.

Lemma 2.3. *The right Kan extension R_* of R_- is the standard FCA closure operator $(-)_\Delta$.*

Proof. The proof is completely analagous to Lemma 2.2, as the right Kan extension has the formula,

$$R_*(B)(g) = \int_{m \in M} \mathbb{2}(B(m), gRm).$$

Hence, the same argument works and we recognize R_* as $(-)_\Delta$. $\square$

These two functors are not unrelated, but form an adjoint pair on the powerset categories.

Lemma 2.4. *The two operators R^* and R_* form an adjoint pair*

$$[G^{\text{op}}, 2] \xrightleftharpoons[R_*]{R^*} [M, 2]^{\text{op}}.$$

Proof. First, as R^* preserves colimits, it has a right adjoint. For this adjoint to be R_* we need to show that for any $A \in [G^{\text{op}}, 2]$ and $B \in [M, 2]^{\text{op}}$, we have $R^*(A) \subseteq B$ if and only if $A \subseteq R_*(B)$.

By the definition of R^* , we have $R^*(A) \subseteq B$ if and only if for all $m \in M$,

$$\bigwedge_{g \in G} (A(g) \implies gRm) \subseteq B(m).$$

In 2 , the join is the universal, hence this is equivalent to the statement that

$$\forall m \in M, \forall g \in G (A(g) \implies gRm) \subseteq B(m).$$

This is again equivalent to

$$\forall g \in G, A(g) \subseteq \forall m \in M (B(m) \implies gRm),$$

which we recognize as

$$\forall g \in G, A(g) \subseteq \bigwedge_{m \in M} (B(m) \implies gRm),$$

which by the definition of R_* is the statement $A \subseteq R_*(B)$, finishing the proof. $\square$

We can now, as is standard in categorical FCA, recognize the formal concepts of (G, M, R) as the fixedpoints of the adjunction $R^* \dashv R_*$, often called the *nucleus* of the profunctor R , see [cite](#).

More formally, the composition $R_* \circ R^*$ is a monad on $P(G)$, and the extents of the formal concepts are the objects in the Eilenberg–Moore category of this monad. Similarly, taking opposite categories gives a monad $R^{*,\text{op}} \circ R_*^{\text{op}}$ on $P(M)$, and its Eilenberg–Moore category gives the intents of the formal concepts. The nucleus consists of pairs of such objects and attributes, i.e., an object in $\text{Nuc}(R)$ is of the form (A, B) where $A = R_*(R^*(B))$ and $B = R^{*,\text{op}}(R_*^{\text{op}}(A))$. By adjunction, however, these determine each other, so we can focus only on one of them, choosing throughout the paper to focus on the object side.

To summarize, the formal concepts arise completely naturally via simple categorical constructions from the profunctor R , via the fixedpoints of the monad $R_* R^*$ on $[G^{\text{op}}, 2]$, defined using right Kan extensions of the two associated currying.

3 Two more operators

Above we defined the functors R^* and R_* as the right Kan extensions of the two curried functors R^- and R_- along the respective Yoneda embeddings. This prompts the idea to also consider the *left* Kan extensions. This will give us another two of the eight operations described by Dubois–Prade.

Lemma 3.1. *Given a formal context $R : G \nrightarrow M$, the left Kan extension $R_!$ of the functor R^- coincides with the possibilistic operator $(-)^{\Pi}$.*

Proof. For the functor R^- , the left Kan extension is defined by the coend formula

$$R_!(A)(m) = \int^{g \in G} (A(g) \otimes gRm).$$

The monoidal structure in 2 is the *and* operation, and as $[M, 2]^{\text{op}}$ is a complete lattice, the coend is the disjunction

$$R_!(A)(m) = \bigvee_{g \in G} (A(g) \wedge gRm).$$

In words, we have $m \in R_!(A)$ if and only if there is some $g \in A$ such that g has the attribute m , which is precisely the definition of the operator $(-)^{\Pi}$. $\square$

Similar to before, we also have the left Kan extension of the other curried functor R_- . But, as one perhaps should expect from [cite](#), this is not $(-)^{\Pi}$ – it is rather the possibilistic operator $(-)_N$.

Lemma 3.2. *The left Kan extension $R^!$ of the functor R_- coincides with the possibilistic operator $(-)_N$.*

Proof. As for $R_!$, the left Kan extension is given by

$$R^!(B)(g) = \int^{m \in M} (B(m) \otimes gRm).$$

$\square$

As for the other pair of functors R^* and R_* , these also form an adjoint pair.

Lemma 3.3. *The left Kan extensions $R_!$ and $R^!$ of the two curried functors R^- and R_- form an adjoint pair*

$$[G^{\text{op}}, 2] \xrightleftharpoons[R^!]{R_!} [M, 2]^{\text{op}}.$$

Proof. Since $R_!$ preserves all joins, it has a right adjoint. For this to be recognized as $R^!$ we need to show that for all $A \in [G^{\text{op}}, 2]$ and $B \in [M, 2]^{\text{op}}$ we have $A \subseteq R^!(B)$ if and only if $R_!(A) \subseteq B$.

By definition, holds if and only if for all $m \in M$ we have $\bigvee_g (A(g) \wedge gRm) \implies B(m)$, which holds if and only if for all $g \in G$ we have that if g is in A , then $\bigwedge_m (gRm \implies B(m))$. This, we recognize as the definition of $(-)_N$, finishing the proof. $\square$

3.1 The complement profunctor

For any boolean profunctor R one can also define its complement $\bar{R}: G^{\text{op}} \times M \longrightarrow 2$, defined by $g\bar{R}m := \neg gRm$. This defines the complement formal context $(G, M, \bar{R})$ as discussed in [cite](#).

As $\bar{R}$ is a boolean profunctor in its own right, it also has two adjoint pairs, $\bar{R}^* \vdash \bar{R}_*$ and $\bar{R}_! \vdash \bar{R}^!$, coming from the left and right Kan extension as in [Lemma 2.4](#) and [Lemma 3.3](#). We claim that these give rise to the remaining four possibilistic operators via a duality between

operators on (G, M, R) and operators on the complement context $(G, M, \bar{R})$.

The negation operator $\neg$ gives a natural duality operator on the presheaf category

$$\neg: [G^{\text{op}}, 2] \longrightarrow [G^{\text{op}}, 2]^{\text{op}}$$

defined by $(\neg A)(g) = \neg(A(g))$ [cite](#). Any presheaf is in our case equivalent to a subset $A \subseteq G$, and the corresponding subset $\neg A$ is precisely all those g not contained in A , also called its *complement*. This also explains the contravariance, as the complement flips the direction of inclusions of subsets. We have a completely analogous duality on $[M, 2]^{\text{op}}$, which we also denote $\neg$ for simplicity.

Taking this negation dual twice is an equivalence, meaning $\neg$ is an *involution*. Given an operator F on $P(G)$, we will call precomposing with the complement operator the *right dual* of F , and denote it $F\neg$. Similarly we define the *left dual* to be $\neg F$.

Lemma 3.4. *The right dual of the operator $\bar{R}^*$ coincides with the possibilistic operator $(-)^N$.*

Proof. By definition, the functor is given pointwise by

$$\bar{R}^*(\neg A)(m) = \bigwedge_{g \in G} (\neg A(g) \implies g\bar{R}m).$$

Using the boolean equivalence

$$P \implies Q \iff \neg P \vee Q,$$

which gives

$$\bar{R}^*(\neg A)(m) = \bigwedge_{g \in G} (A(g) \vee g\bar{R}m).$$

Now, we can apply the fact that $\neg R = \bar{R}$ together with the above boolean equivalence in reverse, which gives

$$\bar{R}^*(\neg A)(m) = \bigwedge_{g \in G} (gRm \implies A(g)),$$

which is the definition of $(-)^N$. $\square$

Similarly, for the Kan extension of the other curried functor $\bar{R}_-$ we obtain the operator $(-)^{\Pi}$.

Lemma 3.5. *The left dual $\neg \bar{R}_*$ of the right Kan extension of the curried functor $\bar{R}_-$, coincides with the possibilistic operator $(-)^{\Pi}$.*

Proof. By definition we have

$$\neg \bar{R}_*(B)(g) = \neg \bigwedge_{m \in M} (B(m) \implies g\bar{R}m),$$

which by the boolean equivalence

$$P \implies \neg Q \iff \neg P \vee \neg Q$$

gives

$$\neg \bar{R}_*(\neg B)(g) = \neg \bigwedge_{m \in M} (k \neg B(m) \vee \neg g\bar{R}m).$$

Using De Morgan duality we have

$$\neg \bar{R}_*(B)(g) = \bigvee_{m \in M} (B(m) \wedge g\bar{R}m),$$

which is the definition of $(-)_\Pi$. $\square$

From this we automatically obtain one of the main results from [cite](#). The categorical viewpoint reveals exactly why this results appears, which was one of our motivations for this paper.

Corollary 3.6. *The fixedpoints of the asymmetric composition $(-)_\Pi \circ (-)^N$ gives the formal concepts of the complement context $(G, M, \bar{R})$.*

Proof. The composition $(-)_\Pi \circ (-)^N$ is by [Lemma 3.4](#) and [Lemma 3.5](#) equivalent to $\neg \bar{R}_* \circ \bar{R}^* \neg$, which takes any subset A to its complement, forms the formal concept – in $[G^{\text{op}}, 2]^{\text{op}}$ the composition of the two functors changes order, and hence really gives the formal concept – and finally takes the complement to the original lattice. The fixedpoints are then precicely the formal concepts in the complement lattice. $\square$

Now we have but two operators left, which arise from the two left Kan extensions associated to the two curryings of $\bar{R}$. The proofs are essentially just using the defintions of the operators.

Lemma 3.7. *The right dual $\bar{R}_! \neg$ of the left Kan extension of the curried functor $\bar{R}^-$, is equivalent to the possibilistic operator $(-)^{\nabla}$.*

Proof. By definition we have

$$\bar{R}_!(\neg A)(m) = \bigvee_{g \in G} (\neg A(g) \wedge g\bar{R}m).$$

which we immediately recognize as the operator $(-)^{\nabla}$. $\square$

Lemma 3.8. *The right dual $\bar{R}^! \neg$ of the left Kan extension of the curried functor $\bar{R}_-$, is equivalent to the possibilistic operator $(-)^{\nabla}$.*

Proof. The proof is the same as [Lemma 3.7](#). $\square$

To summarize then, we have four pairs of adjoint functors, arising from two boolean profunctors. This gives [Table 1](#), where we have denoted operators as either a right Kan extension (Ran) or a left Kan extension (Lan) and omitted the duals.

	Ran	Lan
R^-	$(-)^{\Delta}$	$(-)^{\Pi}$
R_-	$(-)_{\Delta}$	$(-)_{\Pi}$
$\bar{R}^-$	$(-)^N$	$(-)^{\nabla}$
$\bar{R}_-$	$(-)_{\Pi}$	$(-)_{\nabla}$

Table 1: Possibilistic operators as Kan extensions

3.2 Properties

In [cite](#), several properties and relations between the eight possibilistic operators were proven. In this section we reprove these from standard categorical properties. First, however, we prove that this categorical perspective also recovers the De Morgan dualities between these eight operators, as is their usual classical definitions in possibility theory – the necessity is usually defined as $N(A) = 1 - \Pi(\neg A)$, and similarly, $\nabla(A) = 1 - \Delta(\neg A)$.

Lemma 3.9. *The four adjoint pairs of operators are related by De Morgan dualities: $(-)^N = \neg(-)^{\Pi} \neg$ and $(-)^{\nabla} = \neg(-)^{\Delta} \neg$, and similarly for the lower-case operators.*

Proof. We only prove the first relation, as the others follow by symmetry.

Let $A \in [G^{\text{op}}, 2]$ be a presheaf. By definition we have

$$\bar{R}^*(A)(m) = \bigwedge_{g \in G} (A(g) \implies g\bar{R}m).$$

Using $g\bar{R}m = \neg g\bar{R}m$ and the boolean equivalence

$$P \implies \neg Q \iff \neg(P \wedge Q),$$

we have

$$\overline{R}^*(A)(m) = \bigwedge_{g \in G} \neg(A(g) \wedge gRm).$$

This is, by De Morgan duality [cite](#), equivalent to

$$\overline{R}^*(A)(m) = \neg \bigvee_{g \in G} (A(g) \wedge gRm),$$

which we can recognize as $\neg R_!$.

By [Lemma 3.4](#) and [Lemma 3.1](#) we have $(-)^N = \overline{R}^* \neg$ and $(-)^{\Pi} = R_!$, giving

$$(-)^N = \overline{R}^* \neg = \neg R_! \neg = \neg(-)^{\Pi} \neg,$$

finishing the proof. $\square$

The following reproves [cite](#). The proof uses some simple calculus of limits and colimits, which one needs to keep straight as there are a lot of opposite categories.

Proposition 3.10 ([cite](#)). *Let (G, M, R) be a formal context. For all subsets A_1, A_2 of G , we have:*

1. $(A_1 \cup A_2)^{\Delta} = A_1^{\Delta} \cap A_2^{\Delta}$,
2. $(A_1 \cap A_2)^N = A_1^N \cap A_2^N$,
3. $(A_1 \cup A_2)^{\Pi} = A_1^{\Pi} \cup A_2^{\Pi}$, and
4. $(A_1 \cap A_2)^{\nabla} = A_1^{\nabla} \cup A_2^{\nabla}$.

Proof. We prove these one by one in the order presented.

1. As $(-)^{\Delta}: [G^{\text{op}}, 2] \rightarrow [M, 2]^{\text{op}}$ is a left adjoint it preserves colimits, which in $[G^{\text{op}}, 2]$ are unions and in $[M, 2]^{\text{op}}$ are intersections.
2. The functor $(-)^N = \overline{R}^* \neg$ is the dual of a left adjoint. As $\neg: [G^{\text{op}}, 2] \rightarrow [G^{\text{op}}, 2]^{\text{op}}$ is an involution, it preserves both limits and colimits. Hence, intersections, which are limits in $[G^{\text{op}}, 2]$, are sent to colimits in $[M, 2]^{\text{op}}$, which are also intersections.
3. Follows from (2) and the De Morgan duality of [Lemma 3.9](#).
4. Follows from (1) and [Lemma 3.9](#).

$\square$

Similarly, we get the usual inclusion properties of these operators from simple categorical observations.

Proposition 3.11. *Let (G, M, R) be a formal context. For all subsets A of G , we have:*

1. $A \subseteq A_{\Delta}^{\Delta}$
2. $A \subseteq A_N^{\Pi}$
3. $A_{\Pi}^N \subseteq A$
4. $A_{\nabla}^{\nabla} \subseteq A$

Proof. Again we prove these in the order presented.

1. Follows immediately from the unit of the adjunction $\text{Id} \rightarrow (-)_{\Delta} \circ (-)^{\Delta}$.
2. Same as (1), it follows from the adjunction unit of $(-)^{\Pi} \dashv (-)_N$.
3. We have $A_{\Pi}^N = \neg \overline{R}_* \overline{R}^*(A) \neg$, hence this also follows from the adjunction unit of $\overline{R}^* \dashv \overline{R}_*$ via the involution $\neg$.
4. Similar to (3), this follows from the adjunction unit or $\overline{R}_! \dashv \overline{R}^!$.

$\square$

4 Compositions and closures

We have now constructed the eight possibilistic operators in formal concept analysis, as introduced in [cite](#), by simple categorical constructions associated to the underlying boolean profunctor $R: G \nrightarrow G$. It remains then, to study the different possible compositions of upper-case and lower-case possibilistic operators.

Given our eight operators, there are 16 possible compositions that produce an endofunctor on $[G^{\text{op}}, 2]$. As we only consider endofunctors on $[G^{\text{op}}, 2]$ – the ones on $[M, 2]^{\text{op}}$ are completely analagous – we will use the combined notation $(-)^{\sigma} \circ (-)_{\rho} := (-)_{\rho}^{\sigma}$ for simplicity. We then get the following table of compositions:

	$(-)_{\Delta}$	$(-)_{\Pi}$	$(-)_{\nabla}$	$(-)_{\Delta}^{\Delta}$
$(-)_{\Delta}$	$(-)_{\Delta}^{\Delta}$	$(-)_{\Pi}^{\Delta}$	$(-)_{\nabla}^{\Delta}$	$(-)_{\Delta}^{\Delta}$
$(-)_{\Pi}$	$(-)_{\Delta}^{\Pi}$	$(-)_{\Pi}^{\Pi}$	$(-)_{\nabla}^{\Pi}$	$(-)_{\Delta}^{\Pi}$
$(-)_{\nabla}$	$(-)_{\Delta}^{\nabla}$	$(-)_{\Pi}^{\nabla}$	$(-)_{\nabla}^{\nabla}$	$(-)_{\Delta}^{\nabla}$
$(-)_{\Delta}^{\Delta}$	$(-)_{\Delta}^{\Delta}$	$(-)_{\Pi}^{\Delta}$	$(-)_{\nabla}^{\Delta}$	$(-)_{\Delta}^{\Delta}$
$(-)_{\Delta}^{\Pi}$	$(-)_{\Delta}^{\Pi}$	$(-)_{\Pi}^{\Pi}$	$(-)_{\nabla}^{\Pi}$	$(-)_{\Delta}^{\Pi}$
$(-)_{\Delta}^N$	$(-)_{\Delta}^N$	$(-)_{\Pi}^N$	$(-)_{\nabla}^N$	$(-)_{\Delta}^N$
$(-)_{\Delta}^{\nabla}$	$(-)_{\Delta}^{\nabla}$	$(-)_{\Pi}^{\nabla}$	$(-)_{\nabla}^{\nabla}$	$(-)_{\Delta}^{\nabla}$

By our construction of the operators we have two adjunctions associated to the profunctor R . We have the FCA closure adjunction,

$$[G^{\text{op}}, \mathbb{2}] \xrightleftharpoons[(-)_{\Delta}]{(-)^{\Delta}} [M, \mathbb{2}]^{\text{op}}$$

defined by the right Kan extension, and we have the adjunction defined by the left Kan extension,

$$[G^{\text{op}}, \mathbb{2}] \xrightleftharpoons[(-)^N]{(-)^{\Pi}} [M, \mathbb{2}]^{\text{op}}.$$

We also have the two adjunctions associated to the complement profunctor $\bar{R}: G \rightarrow M$:

$$[G^{\text{op}}, \mathbb{2}] \xrightleftharpoons[(-)^{\nabla}]{(-)_{\nabla}} [M, \mathbb{2}]^{\text{op}}.$$

and

$$[G^{\text{op}}, \mathbb{2}] \xrightleftharpoons[(-)^N]{(-)^{\Pi}} [M, \mathbb{2}]^{\text{op}},$$

where due to the use of $\neg$, the adjunctions are flipped after moving to the correct concept lattice.

These latter two, then, form *interior operators*, as we can see from [Proposition 3.11](#). The goal of this section is to prove that the former two are the *only* two of the 16 compositions that form closure operators on $P(G)$.

4.1 Idempotency and closure

By definition, a closure operator is extensive, monotone and idempotent. For an operator F , this means that

1. $A \subseteq F(A)$,
2. $A \subseteq A' \implies F(A) \subseteq F(A')$, and
3. $FF(A) = F(A)$, respectively.

The FCA adjunction coming from a formal context (G, M, R) satisfies all three of these. By [cite](#), the monad R_*R^* is idempotent, and the two other properties follow formally by the adjunction unit and the fact that it is a functor on posets. Thus, any monadic functor on a poset satisfies these, see [cite](#).

In order to be recognized as a FCA closure operator on *some* formal context, these three axioms have to be fulfilled, otherwise the resulting fixedpoints are not formal concepts. The following is rather trivial then, but we state it for reference later.

Proposition 4.1. *Let $L: \mathcal{C} \rightarrow \mathcal{C}$ be an endofunctor on a poset. If L fails to be extensive, monotone or idempotent, then there is no formal context (G, M, R) such that $\text{Fix}(L) \simeq \text{Nuc}(R)$.*

This result will help us quickly identify which of the asymmetric compositions later possibly could give rise to alternative formal concepts on some related formal context. However, satisfying these three properties does not, as far as we know, guarantee that the fixed-points form the formal concept of some formal context.

Remark 4.2. Dually, given an adjoint pair $L \dashv R$, the composition LR is a comonad. On posets, this comonad satisfies dual properties of the closure operators, called being an *interior operator*. In the case of formal concepts, instead of completing a subset A to a formal concept, i.e., constructing the smallest formal concept containing the set, the interior operator finds the largest formal concept *inside* A .

4.2 Binary compositions

We can now classify exactly which binary compositions of the eight operators that give closure operators and interior operators.

Theorem 4.3. *If F is an operator in table X, then F is a closure operator only if it is the monad of one of the four adjunctions.*

Proof. The monads of the adjunctions are always closure operators, but due to the use of $\neg$, two of them are in effect interior operators. In particular, $(-)_{\Delta}^{\Delta}$ and $(-)_{\Pi}^N$ are closure operators, while $(-)_{\Pi}^{\nabla}$ and $(-)_{\nabla}^{\nabla}$ are interior operators – see [Proposition 3.11](#). Hence, we only need to check the remaining 12 cases. We check these by simple counter examples, using [Proposition 4.1](#) to test whether F satisfies the properties. In fact, it will be enough to test whether F is extensive.

First, let $G = \{a, b\}$, $M = \{x, y\}$ and R given by aRx, bRy . Chose the subset $A = G$. Then we have $A^{\Delta} = \emptyset$, $A^{\Pi} = \{x, y\}$, $A^N = \emptyset$ and $A^{\nabla} = \emptyset$. In this example, both $A_{\Pi}^{\Delta} = \emptyset$ and $A_N^{\Delta} = \emptyset$, showing that neither operators $(-)_{\Pi}^{\Delta}$ or $(-)_{N}^{\Delta}$ are closure operators. Similarly, we get $A_{\Delta}^{\Pi} = \emptyset$, $A_{\nabla}^{\Pi} = \emptyset$, $A_N^N = \emptyset$, $A_{\nabla}^N = \emptyset$, $A_{\Pi}^{\nabla} = \emptyset$ and $A_N^{\nabla} = \emptyset$, showing that none of these can be closure operators as well.

It remains to show the same for $(-)_\Pi^\Pi$, $(-)_\nabla^\Delta$, $(-)_N^N$ and $(-)_\Delta^\nabla$. For the former two we use the formal context $G = \{a\}$, $M = \emptyset$ and $A = G$. This gives $A_\Pi^\Pi = \emptyset$ and $A_\nabla^\Delta = \emptyset$, showing neither is extensive and hence not a closure operator. Similarly, using $G = \{a\}$, $M = \{x\}$, aRx and $A = G$ $A_\Delta^\nabla = \emptyset$. For the final operator, $(-)_N^N$ we use the context $G = \{a, b\}$, $M = \{x\}$, aRx , bRx and $A = \{a\}$. This gives $A_N^N = \emptyset$, which also finishes the proof. $\square$

We round off this section by using the categorical machinery to identify a sequence of new types of concepts, or pairs, that naturally arise from the framework. These come from the following observation: Given two adjunctions

$$\mathcal{C} \xrightleftharpoons[R_1]{L_1} \mathcal{D} \xrightleftharpoons[R_2]{L_2} \mathcal{E}$$

the composed functor $F_2 \circ F_1$ is a left adjoint to $G_2 \circ G_1$. Recall from [Proposition 4.1](#) that on complete lattices, closure operators are equivalent to monads.

On $P(G)$ we have two left adjoints, $(-)_\Delta^\Delta$ and $(-)_\Pi^\Pi$. Similarly, on $P(M)^{\text{op}}$ we also have two left adjoints, $(-)_\Pi$ and $(-)_\nabla$. Hence, any arbitrary composition of these will produce a closure operator on $P(G)$ by composing with the corresponding adjoints.

Example 4.4. The functor

$$(-)_\Delta^\Delta \circ (-)_\Pi \circ (-)_\Delta^\Delta: [G^{\text{op}}, \mathbb{2}] \longrightarrow [M, \mathbb{2}]^{\text{op}}$$

is a composition of left adjoints, hence is a left adjoint itself with right adjoint given by

$$(-)_\Delta \circ (-)^N \circ (-)_\Delta: [M, \mathbb{2}]^{\text{op}} \longrightarrow [G^{\text{op}}, \mathbb{2}].$$

The composition of these is by [Proposition 4.1](#) a closure operator.

We believe that the corresponding concepts of these closure operators would be interesting to study, as they are truly *new* types of concepts.

5 Conclusion

We have used simple categorical ideas to construct the eight possibilistic operators introduced by Dubois-Prade in [cite](#). We used this to identify some of their properties as arising from abstract nonsense, as well as

classified exactly which binary compositions, asymmetric or symmetric, which give closure operators. Finally we introduced infinitely many new closure operators on $P(G)$ by using ideas from category theory, none of which have been studied in the literature as far as we know.

The categorical approach open up a new avenue of research, and a new way of understanding existing results. In [cite](#), the authors extend the eight possibilistic operators in formal concept analysis to its fuzzy sibling. These results can be obtained in this categorical framework by replacing the boolean category $\mathbb{2}$ by a fuzzy lattice L . More generally, by enriching in an epistemic uncertainty framework, as introduced in our previous work [cite](#), we can obtain a plethora of extensions to this paper into enriched, uncertain settings. We believe this to be a fruitful direction for future work.